Broadcasting Weather Report to Vehicles in Urban Roads

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***Abstract*—Vehicular ad hoc network is widely used to broadcast emergency, weather and road conditions among vehicles. In this paper, an algorithm is proposed to transmit weather information to the vehicles which are travelling in urban areas. The system uses a set of algorithms and rules to determine dangerous weather situation on roads. It is proposed to use Weather_Forecast algorithm to predict the weather conditions on urban areas. The performance and the results of the proposed system are also discussed.**

Keywords—Weather prevention, wireless sensor networks, VANETs, safety messages.

I. INTRODUCTION

Vehicular ad hoc networks (VANETs) is a special form of mobile ad hoc networks (MANETs) which is widely used for exchanging useful information (e.g. comfort and safety messages). VANET is an emerging technology that enhance smooth driving for passengers by transmitting the road conditions and weather information in advance.

VANET integrates ad hoc networks, wireless communication and sensor technology to achieve intelligent vehicular communication to improve road traffic safety and efficiency. Vehicles today are equipped with wireless communication functions that can facilitate vehicle-to- vehicle and vehicle-to-infrastructure communication. Transmission of weather information to vehicles may help to avoid major accidents. The Intelligent Transportation System (ITS) provides various facilities to exchange such information between vehicles. Vehicles can receive the messages from Road Side Units (RSUs) and may forward such messages to its neighbouring vehicles.

Increased storage capacity, computing and communications power together with advances in wireless networking technology have enabled a host of new applications to improve driving comfort and passenger safety in vehicles. VANET can support for two types of communication: Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication.

Wireless connectivity and special sensors deployed in vehicles and on highways are utilized to continuously report real-time traffic and environmental data. Advance reception of weather information may enable the drivers to take diversion thereby reduces wasting more time in critical road which has adverse weather condition. One of the primary objectives of comfort applications are to send the road weather information to the vehicles. In order to achieve this, the road weather information will be received from the weather station would be sent to RSUs. RSUs in turn forward this information to the vehicles that are travelling in its communication range.

The comfort applications enhance the comfort level of drivers and passengers by disseminating optimal route, traffic related information, weather conditions, location of gas station, nearest restaurants and price information or interactive communication such as Internet access and downloads, entertainment and advertisements.

The weather stations are generally placed in hill or higher areas. The accurate information about weather details can get only from those places. These information later would be sent to the data centres that are available on road sides.

The IEEE 802.11p VANET standard is used as the primary communication with traditional Wi-Fi. The wireless equipment available on each vehicles help to communicate with RSUs that are mounted on road sides.

This paper is organized as follows: Section II depicts the related work. Section III describes the proposed system. Section IV depicts the results and discussions. Section V outlines the conclusion.

II.RELATED WORK

VANET are becoming an advanced technology to improve in terms of safety, comfort and efficient. Climatic change is one of the major issues for drivers and passengers. Advance receipt of weather information may help the drivers to find an alternate path in order to reach their designated destination. Traffic management takes a vital role in improving traffic safety under adverse weather condition and to develop a traffic

prediction algorithm to reduce the effect of rainfall and other unbearable weather situations.

The road conditions such as ambient conditions (rainfall, flood, snow) traffic in the road condition (emergency, construction sites) etc. The main aim of this paper is to transmit the weather condition to vehicles through RSUs. The weather information can be forwarded to vehicles using vehicle to vehicle communication (v2v) and vehicle to road infrastructure (V2I) communication.

Vehicles become damage due to percentage of paves roads over 50% rainfall, 20% snowfall, 15% ice pavement,15% sunshine. Adverse weather conditions affect directly driving conditions and rain reduce visibility but also increases the risk of slippery roads and affects stability of the vehicle more over braking distance increases. In this paper, we are going to present a new expert system to identify local road network [1].

Road safety can be improved providing real-time local information to the road users, especially during first period of the adverse situation. Traffic management in the real time is a complex task since in high level of distribution. due to heavy rain in the city water level in road raises that leads to puddle, water logging across street, and create traffic problem. road have a huge extension and different equipment's are used to monitor the traffic status, support to control, manage to the traffic situation. These systems are known as Advanced Traffic Management System (ATMS) could improve the road safety [2]. Climate Data Online (CDO) provide free access to National Climatic Data Centre (NCDC) which provide historical weather and climatic information. These data include disastrous weather condition controlled daily, monthly, seasonally and yearly measurements of temperature precipitation, wind and degree as well as radar data for 300 year's climate conditions [3]. Nowadays multiple computing techniques are available which can be used for forecasting accuracy and there are different forecasting methods are Naïve Approach Judgmental Methods, Quantitative and Qualitative Method, Econometric or Casual Forecasting Methods, Time Series Methods, Artificial Intelligence Methods, etc. The weather forecasting is live forecasting where output of the model may be required for daily weather guide or weekly or monthly weather plans. Thus, accuracy of the result is very important aspect in forecasting [3].

The authors Stephen Dunne et al have been proposed a weather prediction algorithm to forecast hourly traffic flow under worst weather condition. They could get accurate predictions on two urban arterial location like Bangalore, Mysore [4]. They have planned to extend their work to forecast from neuro-wavelet model to standard artificial neural network model.

The authors Martin Berne et al have been described weather warning system on roads. They have analysed the rapidly changing weather and vast network infrastructure which cover's local weather stations, regional and national management centre. They have also mentioned that the response time taken to connect network with system integration was from 5 to 10 minutes [5]. They have also planned to extend their work to include security issues and traffic influence with the system consideration.

The authors Timo Sukuvaara et al have been proposed a system to check winter conditions of road traffic platform in the northern Scandinavia [5]. They have planned to extend their work to generate added value in terms of safety and convenience in any geographical area experiencing harsh winter conditions.

The authors Yella Siril, Kalid Askar and Mark Dougherty have been proposed a system to disseminate information such as slipperiness on the road. They have also proposed an approach for predicting the extend of slipperiness by building and testing an expert system [6].

The authors Anita Raja et al have been proposed an embedded system which consists of collaborating agents that are capable of interacting with their environments [7]. They have used Decentralized Markov Decision Process (DEC-MDP) for decision making in tracking application. They have also pointed out that the cost of meta-level control is high for constructing and evaluating the DEC-MDPs offline.

The authors Joachim Wahel, Michael Schreckenberg et al have been proposed an online simulation and predictions which are based on combination of real-world traffic data and a multi-agent traffic flow model. It includes decision-making and route choice behaviour of the road users [8].

The authors Petr Kalina et al have been proposed a vehicle routing problem to route weather related information with time based on agent negotiation [9]. The authors Meera Narvekar bnd Priyanca Fargose have been proposed a daily weather forecasting approach in multiple areas like agriculture, fisherman, transportation etc. [3].

The authors Vicente.R, Tomas, Marta have been proposed a new autonomous system for broadcasting weather conditions to vehicles [10]. They used some set of algorithms and rules for determining the dangerous condition of road in urban areas. They have also planned to extend their work to collect the information from various stations and the predictions to be done in synchronized manner.

The author Vicente.R, Tomas, Marta have been proposed the important factor of weather which affect traffic flow and road safety and also support road operators to inform the safety information to drivers [9]. They have used High Resolution Limited Area Model (HIRLAM) that emerged with the aim of providing a short-term weather prediction with acceptable reliability. They have developed with the cooperation of European meteorological institutes [10]. Finally, the authors have compared all the techniques with their accuracy and planned to develop a new model to produce accurate results.

III. PROPOSED SYSTEM

The aim of proposed system is to reduce traffic problem and provide safe zone for drivers and people even in dangerous weather situation. This can be achieved by integrating various devices such as sensors, controllers, communication devices etc. The V2V and V2R or V2I supports for disseminating weather information and it is able to collect and exchange data among the drivers through On Board Unit (OBU) and road side infrastructure enabled, probably in street lights. The objectives of the proposed systems are listed below:

- In proposed system, we are going to present a new system to identify adverse weather situation using Weather_Forecast algorithm in a local road network.

  The objective of this system is to provide short term prediction on unfavourable weather conditions for the next 24 hours with a high degree of reliability.

The objective of this is to provide short term weather predictions and provide safe traffic free zone for drivers and people. Figure 1 shows the architecture of the proposed system. This algorithm helps to predict what happens in next 24 hours by comparing and detecting dangerous situations happened in past years using days and times. It is assumed that the vehicles can view the historical weather report from various location connected to that particular server which is connected to the online weather services called Climate Data Online (CDO) . CDO provides free access to National Climate Data Centre's that gives a global historical weather and climatic data. In general, weather report includes daily, monthly, seasonally and yearly measurement of temperature in the form of liquid or solid. Authority will have a collection of historical data for the past 2 to 5 years. This data is used by the data administrator to compare and predict by city, location, day, time, seconds and exact queries of what going to happen in next 24 hours in that particular location. Authority will send alert message to RSU including message about normal weather report and dangerous disaster emergency message. OBU can able to receive alert message with in short duration.

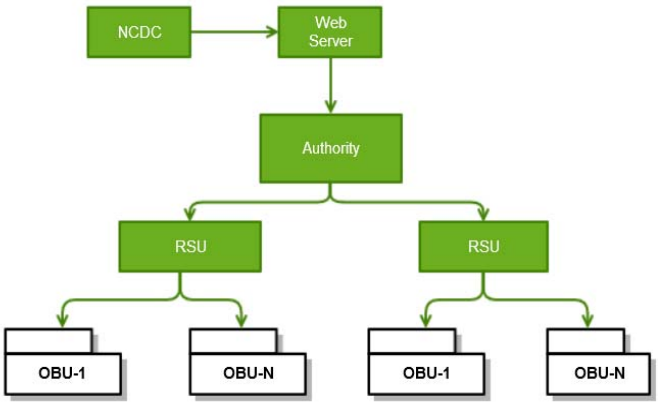

Figure 1: Weather Forecasting Architecture

Authority will have details of road side network by using mobile sensor range which consists of moving and non-moving vehicles. A network will have different parts of region and each region contains N-number of nodes. Authority will have both data of weather database and road situation condition. Authority will combine both the information and send to web server from there it will send to other relevant area's so, any vehicle coming to the affected place can easily predict and send them alert message before any dangerous threshold happened drivers or people on roads. The key operations of this proposed system is described below:

OPERATIONS:

1. The RSUs receive weather information the weather center located in the region.

2. The RSUs send the hello beacon messages including weather condition periodically to vehicles.

3. After receiving the beacon packets by the vehicles, the warning messages will be brought to the driver's knowledge in the form of text or audio sound.

MODULES

- **NETWORK FORMATION**

A network is first formed with different regions which are divided based on the mobile sensor range. Each node sends "HELLO" message to same interest nodes. Once the node detects "HELLO" message from another node (neighbour) then network is formed with different regions. Regions are divided based on mobile sensor ranges.

- **V2V COMMUNICATION THROUGH RSU**

In V2V communication, vehicles can communicate through the region. The weather information generated by authority will be sent to RSUs. Before transmit the data to vehicles, RSU will verify it and send to all OBUs using V2V, V2I communication mechanisms and is able to collect and exchange data among the drivers, OBU and roadside infrastructure.

- **WEB SERVER PROCESS:**

Road Weather Information Sensor (RWIS) which include visibility, wind speed, temperature, humidity. A data collection station (DCS) which collects traffic data from the road network like flow, speed, density [9]. we send a VMS (variable message signal) which shows message to the road network users. Broadcast weather warning message (daily) to RSU monitor the delayed OBU duration based on validated information.

WEATHER PREDICTION RULES:

Weather forecasting is the application of science and technology to predict temperature, humidity, location. Broadcasting historical emergency weather situation to vehicle is a key point in our proposed system.

ALGORITHMS USED:

In other existing approaches, the weather conditions were identified using number of algorithms like detection algorithm, certainty algorithm, consistency algorithm, correlation algorithm, forecasting algorithm, historical algorithm. But in the proposed system, a single Weather_Forecast algorithm is used to replace all the aforementioned six algorithms.

Weather_Forecast Algorithm:

Weather forecasting are done by collecting quantitative data from current state to previous state for atmosphere predicting and how atmosphere will change. These predictions are made by humans based on clouds changing, current weather condition, barometric pressure, sky condition.

The main objective of our proposed system is to identify the dangerous situation in accurate way for that weather prediction algorithm is used. The main intension of this algorithm is to fetch information, gather information, identify, detect, provide accurate prediction in short term. Weather_Forecast algorithm fetches all information from NCDC (national climate data centre) using CDO (climate data online) web access page. Then, all the information is gathered based on year-month-date-min-sec. Web server is used to monitor current climate database.

Weather database that establishes actual occurrence of each situation and have a numeric calculation between 0 to 100. If the value lies between 0 to 10, then particular situations is not possible else if the value lies between 10 to 100 then completely true situations. By using these detection, we can predict complete scenario of weather that what will happen in next two hours in short period of time. Weather prediction will be more accurate because in historical database we have past 300 years of weather database by using current climatic weather, temperature and cloud condition estimations are done using time, minutes, seconds based on this from past data are listed and calculated by using average of all data so, the predictions are made preventions are done.

IV.RESULT AND DISCUSSION

This paper discusses broadcasting historical weather report to vehicle in urban roads using traffic, road condition and weather information. There is a new system used to forecast weather condition in short time with more accuracy of result. The system uses a new algorithm for identifying and detecting dangerous situation which provide accurate prediction of weather on road network. So, we can predict unexpected increase or decrease rainfall, flood, road damage intensity and events of climatic changes can be prevented.

City	Places	Max humidity	Min humidity	Rain	date
Chennai	*Mt road*	*85%*	65%	*20^{0}C*	*12.09,2014*
Madurai	*Ki road*	*83%*	*66%*	*35^{0}C*	*09.04.2013*
Korai	*Sky salami*	*77%*	*58%*	*24^{0}C*	*02.02.2015*
Ooty	*Mount pct.*	*89%*	*55%*	*28^{0}C*	*24.07.2013*
Salem	Lk road	90%	75%	24°C	29.05.2011

[a] Table for weather database for past years. .[b] City and date for reference

Table 1: Weather Prediction Module*(database)*

Weather and climatic changes are described in Table-1. The maximum, minimum, normal and constant temperature, humitity of rain were obtained from the COD database based on time and seconds using the proposed algorithm. So, we can determine unexpected increase or decrease or constant rainfall that in turn may cause road damage, flood events.

Weather prediction model Figure 2 describes pressure, visiblity, humidity, temperature of rainfall, flood, precipitation using Weather_Forecast algorithm where prediction will be more accurate and calculation are made based on the previous records from past data using historical database.

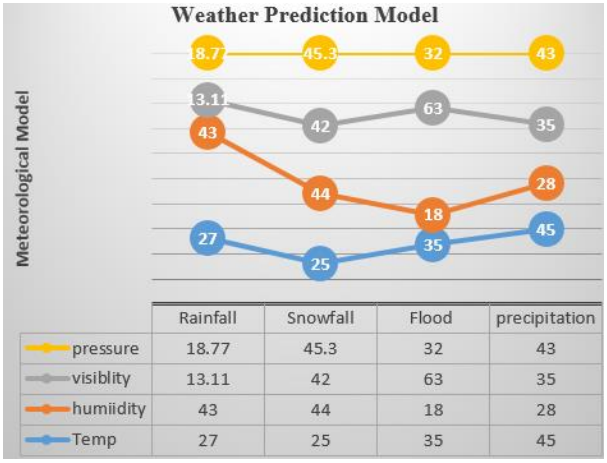

Figure 2: Weather Prediction Model

The weather forcasting has a big challenge of predicing the accurate result which are used in many real time systems like fishery department, tourism centers, airport, metro water department, electricity department etc[8]. Emergency warning is issured before disaster commenced. This is used to alert the people on the relavent occurrence subsequently. Meteorologists are able to predict the analysis and change in weather using various tools and tecnologies. These tools are used to measure atmospheric conditions that occures in past and present. Figure 3 shows the performance analysis of the proposed system.

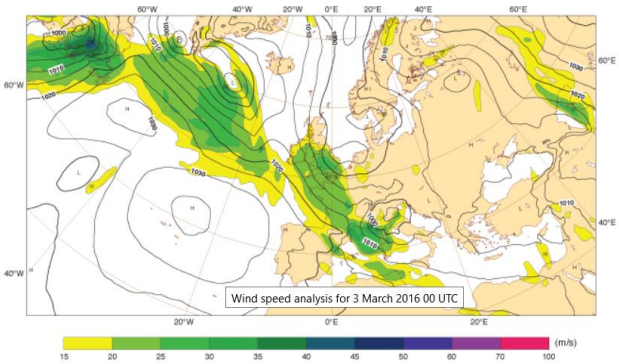

Figure3:Forcasting Performance Analysis Result

Forcasting can be more effectively done using satellites observation. Figure 4 describes daily updates of sunrise-sunset,UV radiation report, wind speed, precipitation and humidity information using historical weather database from various places those information can be fetched from COD which exact and accurate degree of weather.

DAY	DESCRIPTION	HIGH/LOW	PRECIP	WIND	HUMIDITY	UV INDEX	SUNRISE
TODAY OCT 25	Few Showers	57°/50°	30%	SE 9 mph	88%	2 of 10	7:25 am
WED OCT 26	Sunny	63°/50°	0%	SSE 11 mph	62%	4 of 10	7:26 am
THU OCT 27	Rain	55°/49°	100%	ESE 7 mph	81%	2 of 10	7:28 am
FRI OCT 28	Rain	52°/48°	100%	SSE 13 mph	95%	2 of 10	7:29 am
SAT OCT 29	Rain	52°/47°	70%	SE 8 mph	86%	2 of 10	7:30 am
SUN OCT 30	Rain	48°/43°	90%	SSE 12 mph	94%	2 of 10	7:31 am
MON OCT 31	Showers	48°/42°	50%	SSE 7 mph	91%	3 of 10	7:32 am

Figure 4: Historical Database Information

Climatic changes will be in our daily life and there will not be a moderate assumtion of weather.

V.CONCLUSTION

This paper presents a new autonomous system to forecast weather conditions in a short time and to broadcast that information to people through vehicles. The system uses a set of algorithm and rules to determine the weather and to forecast dangerous situations in the road network. It has been implemented using a multi-agent approach and tested with set of data.

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